

ENGR 102 Summer Bridge

Day 8 Student Quiz

Debugging, Boundary Tests, and Complete Repair Evidence

20 points • approximately 20-25 minutes

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Boolean-operator convention. Python operators appear as **and**, **or**, and **not**. Their lowercase spelling is required in code.

Show intermediate state for trace credit. Program credit requires one coherent solution. Unless a prompt says otherwise, give exact Python 3 output.

Question 1. Diagnose an inclusive-boundary defect.

5 points

```
temperature = 35
ready = temperature > 15 and temperature < 35
print(ready)
```

The requirement includes 15 and 35. Give actual result, cause, corrected assignment, and two tests that expose the defect.

Question 2. Build a minimum protective boundary set.

5 points

```
reading = 10
low = 10
high = 20

if reading < low:
    status = "below"
elif reading > high:
    status = "above"
else:
    status = "within"

print(status)
```

Give exact supplied output, then specify four tests that protect both inclusive boundaries and their immediate outside cases.

Question 3. Program construction

10 points

Write one complete Python program that satisfies every requirement.

- Read reading, low, and high as floats.
- Print invalid when low is greater than high.
- Otherwise print Result: below, Result: within, or Result: above. Equality at either limit is within.
- Use one mutually exclusive chain and provide four boundary tests with expected output.

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Test evidence / expected result